Module5

Exercise 2

Part 1: INNER JOIN

1. Retrieve a list of all orders along with the customer's company name.

A1.) Query & Output

Query Query History

```
select c.company_name,o.order_id,o.customer_id from orders o
inner join customers c on c.customer_id=o.customer_id;
```

Data Output Messages Notifications

	company_name character varying (40)	order_id smallint	customer_id character
1	Vins et alcools Chevalier	10248	VINET
2	Toms Spezialitäten	10249	TOMSP
3	Hanari Carnes	10250	HANAR
4	Victuailles en stock	10251	VICTE
5	Suprêmes délices	10252	SUPRD
6	Hanari Carnes	10253	HANAR
7	Chop-suey Chinese	10254	CHOPS
8	Richter Supermarkt	10255	RICSU
9	Wellington Importadora	10256	WELLI
10	HILARION-Abastos	10257	HILAA
11	Ernst Handel	10258	ERNSH
12	Centro comercial Moctezuma	10259	CENTC
13	Ottilies Käseladen	10260	OTTIK

2. Get the order details (OrderID, ProductName, UnitPrice, Quantity) by joining the Orders and OrderDetails tables.

A2.) Query & Output

Exercise2/postgres@localhost

No limit

Query | Query History

```
select o.order_id,p.product_name,od.unit_price,od.quantity
from orders o inner join
order_details od on o.order_id=od.order_id
inner join products p on p.product_id = od.product_id;
```

	order_id smallint	product_name character varying (40)	unit_price real	quantity smallint
1	10248	Queso Cabrales	14	12
2	10248	Singaporean Hokkien Fried Mee	9.8	10
3	10248	Mozzarella di Giovanni	34.8	5
4	10249	Tofu	18.6	9
5	10249	Manjimup Dried Apples	42.4	40
6	10250	Jack's New England Clam Cho...	7.7	10
7	10250	Manjimup Dried Apples	42.4	35
8	10250	Louisiana Fiery Hot Pepper Sau...	16.8	15
9	10251	Gustaf's Knäckebröd	16.8	6
10	10251	Ravioli Angelo	15.6	15
11	10251	Louisiana Fiery Hot Pepper Sau...	16.8	20
12	10252	Sir Rodney's Marmalade	64.8	40
13	10252	Geitost	2	25
14	10252	Camembert Pierrot	27.2	40
15	10253	Gorgonzola Telino	10	20
16	10253	Chartreuse verte	14.4	42
17	10253	Maxilaku	16	40
18	10254	Guaraná Fantástica	3.6	15
19	10254	Pâté chinois	19.2	21
20	10254	Longlife Tofu	8	21
21	10255	Chang	15.2	20
22	10255	Pavlova	13.9	35
23	10255	Inlagd Sill	15.2	25

Total rows: 2155 | Query complete 00:00:00.106

3. Fetch all employees and their respective territories from the Employees and EmployeeTerritories tables.

A.) Query & Output

Exercise2/postgres@localhost

No limit

Query | Query History

```
select e.employee_id,e.first_name,e.region,et.territory_id,t.territory_description
from employees e inner join
employee_territories et on e.employee_id=et.employee_id
inner join territories t on t.territory_id = et.territory_id;
```

Data Output | Messages | Notifications

	employee_id smallint	first_name character varying (10)	region character varying (15)	territory_id character varying (20)	territory_description character
1	1	Nancy	WA	06897	Wilton
2	1	Nancy	WA	19713	Neward
3	2	Andrew	WA	01581	Westboro
4	2	Andrew	WA	01730	Bedford
5	2	Andrew	WA	01833	Georgetow
6	2	Andrew	WA	02116	Boston
7	2	Andrew	WA	02139	Cambridge
8	2	Andrew	WA	02184	Braintree
9	2	Andrew	WA	40222	Louisville
10	3	Janet	WA	30346	Atlanta
11	3	Janet	WA	31406	Savannah
12	3	Janet	WA	32859	Orlando
13	3	Janet	WA	33607	Tampa
14	4	Margaret	WA	20852	Rockville
15	4	Margaret	WA	27403	Greensboro
16	4	Margaret	WA	27511	Cary
17	5	Steven	[null]	02903	Providence
18	5	Steven	[null]	07960	Morristown
19	5	Steven	[null]	08837	Edison
20	5	Steven	[null]	10019	New York
21	5	Steven	[null]	10038	New York
22	5	Steven	[null]	11747	Mellvile
23	5	Steven	[null]	14450	Fairport

Total rows: 49 | Query complete 00:00:00.114

Part 2: LEFT JOIN

1. Retrieve a list of all customers and their orders, including those who haven't placed any orders.

A.) Query & Output

Exercise2/postgres@localhost

No limit

Query | Query History

```
select c.customer_id,c.contact_name,o.order_id
from customers c left join
orders o on c.customer_id=o.customer_id;
```

Data Output | Messages | Notifications

	customer_id character	contact_name character varying (30)	order_id smallint
810	NORTS	Simon Crowther	11057
811	BLAUS	Hanna Moos	11058
812	RICAR	Janete Limeira	11059
813	FRANS	Paolo Accorti	11060
814	GREAL	Howard Snyder	11061
815	REGGC	Maurizio Moroni	11062
816	HUNGO	Patricia McKenna	11063
817	SAVEA	Jose Pavarotti	11064
818	LILAS	Carlos González	11065
819	WHITC	Karl Jablonski	11066
820	DRACD	Sven Ottlieb	11067
821	QUEEN	Lúcia Carvalho	11068
822	TORTU	Miguel Angel Paolino	11069
823	LEHMS	Renate Messner	11070
824	LILAS	Carlos González	11071
825	ERNSH	Roland Mendel	11072
826	PERIC	Guillermo Fernández	11073
827	SIMOB	Jytte Petersen	11074
828	RICSU	Michael Holz	11075
829	BONAP	Laurence Lebihan	11076
830	RATTC	Paula Wilson	11077
831	PARIS	Marie Bertrand	[null]
832	FISSA	Diego Roel	[null]

Total rows: 832 | Query complete 00:00:00.090

2. Display a list of products along with the total quantity sold, including products that have never been ordered.

A.) Query & Output

Query Query History

```sql
select p.product_id, p.product_name, coalesce(sum(od.quantity),0) as total_quantity_sold
from products p
left join order_details od on p.product_id = od.product_id
group by p.product_id, p.product_name;
```

Data Output Messages Notifications

	product_id [PK] smallint	product_name character varying (40)	total_quantity_sold bigint
1	74	Longlife Tofu	297
2	54	Tourtière	755
3	29	Thüringer Rostbratwurst	746
4	71	Flotemysost	1057
5	4	Chef Anton's Cajun Seasoning	453
6	68	Scottish Longbreads	799
7	34	Sasquatch Ale	506
8	51	Manjimup Dried Apples	886
9	52	Filo Mix	500
10	70	Outback Lager	817
11	67	Laughing Lumberjack Lager	184
12	63	Vegie-spread	445
13	10	Ikura	742
14	35	Steeleye Stout	883
15	45	Rogede sild	508
16	6	Grandma's Boysenberry Spread	301
17	39	Chartreuse verte	793
18	69	Gudbrandsdalsost	714

Total rows: 77 Query complete 00:00:00.138

3. Get all suppliers and their products, including suppliers that don't currently supply any products.

A.) Query & Output

Exercise2/postgres@localhost

No limit

Query | Query History

```
select s.supplier_id,s.contact_name,p.product_id,p.product_name
from suppliers s left join
products p on s.supplier_id=p.supplier_id;
```

Data Output | Messages | Notifications

	supplier_id smallint	contact_name character varying (30)	product_id smallint	product_name character varying (40)
1	8	Peter Wilson	1	Chai
2	1	Charlotte Cooper	2	Chang
3	1	Charlotte Cooper	3	Aniseed Syrup
4	2	Shelley Burke	4	Chef Anton's Cajun Seasoning
5	2	Shelley Burke	5	Chef Anton's Gumbo Mix
6	3	Regina Murphy	6	Grandma's Boysenberry Spread
7	3	Regina Murphy	7	Uncle Bob's Organic Dried Pears
8	3	Regina Murphy	8	Northwoods Cranberry Sauce
9	4	Yoshi Nagase	9	Mishi Kobe Niku
10	4	Yoshi Nagase	10	Ikura
11	5	Antonio del Valle Saave...	11	Queso Cabrales
12	5	Antonio del Valle Saave...	12	Queso Manchego La Pastora
13	6	Mayumi Ohno	13	Konbu
14	6	Mayumi Ohno	14	Tofu
15	6	Mayumi Ohno	15	Genen Shouyu
16	7	Ian Devling	16	Pavlova
17	7	Ian Devling	17	Alice Mutton
18	7	Ian Devling	18	Carnarvon Tigers
19	8	Peter Wilson	19	Teatime Chocolate Biscuits
20	8	Peter Wilson	20	Sir Rodney's Marmalade
21	8	Peter Wilson	21	Sir Rodney's Scones
22	9	Lars Peterson	22	Gustaf's Knäckebröd
23	9	Lars Peterson	23	Tunnbröd

Total rows: 77 | Query complete 00:00:00.114

Part 3: RIGHT JOIN

1. Retrieve a list of all orders along with the customer's contact name, ensuring all orders are included, even if the customer’s contact name is missing.

A.) Query & Output

Exercise2/postgres@localhost

No limit

Query | Query History

```
select o.order_id,c.customer_id,c.contact_name
from customers c right join
orders o on o.customer_id=c.customer_id;
```

Data Output | Messages | Notifications

	order_id smallint	customer_id character	contact_name character varying (30)
1	10248	VINET	Paul Henriot
2	10249	TOMSP	Karin Josephs
3	10250	HANAR	Mario Pontes
4	10251	VICTE	Mary Saveley
5	10252	SUPRD	Pascale Cartrain
6	10253	HANAR	Mario Pontes
7	10254	CHOPS	Yang Wang
8	10255	RICSU	Michael Holz
9	10256	WELLI	Paula Parente
10	10257	HILAA	Carlos Hernández
11	10258	ERNSH	Roland Mendel
12	10259	CENTC	Francisco Chang
13	10260	OTTIK	Henriette Pfalzheim
14	10261	QUEDE	Bernardo Batista
15	10262	RATTC	Paula Wilson
16	10263	ERNSH	Roland Mendel
17	10264	FOLKO	Maria Larsson
18	10265	BLONP	Frédérique Citeaux
19	10266	WARTH	Pirkko Koskitalo
20	10267	FRANK	Peter Franken
21	10268	GROSR	Manuel Pereira
22	10269	WHITC	Karl Jablonski
23	10270	WARTH	Pirkko Koskitalo

Total rows: 830 | Query complete 00:00:00.116

2. Fetch all orders and their shippers, ensuring all orders are included, even if no shipper is assigned.

Query Query History

```
select o.order_id,o.order_date,o.ship_via,s.shipper_id,s.company_name as shipper_name,s.phone
from orders o
left join shippers s
on o.ship_via = s.shipper_id
```

Data Output Messages Notifications

	order_id smallint	order_date date	ship_via smallint	shipper_id smallint	shipper_name character varying (40)	phone character varying (24)
1	10248	1996-07-04	3	3	Federal Shipping	(503) 555-9931
2	10249	1996-07-05	1	1	Speedy Express	(503) 555-9831
3	10250	1996-07-08	2	2	United Package	(503) 555-3199
4	10251	1996-07-08	1	1	Speedy Express	(503) 555-9831
5	10252	1996-07-09	2	2	United Package	(503) 555-3199
6	10253	1996-07-10	2	2	United Package	(503) 555-3199
7	10254	1996-07-11	2	2	United Package	(503) 555-3199
8	10255	1996-07-12	3	3	Federal Shipping	(503) 555-9931
9	10256	1996-07-15	2	2	United Package	(503) 555-3199
10	10257	1996-07-16	3	3	Federal Shipping	(503) 555-9931
11	10258	1996-07-17	1	1	Speedy Express	(503) 555-9831
12	10259	1996-07-18	3	3	Federal Shipping	(503) 555-9931
13	10260	1996-07-19	1	1	Speedy Express	(503) 555-9831
14	10261	1996-07-19	2	2	United Package	(503) 555-3199
15	10262	1996-07-22	3	3	Federal Shipping	(503) 555-9931
16	10263	1996-07-23	3	3	Federal Shipping	(503) 555-9931
17	10264	1996-07-24	3	3	Federal Shipping	(503) 555-9931
18	10265	1996-07-25	1	1	Speedy Express	(503) 555-9831

Total rows: 830 Query complete 00:00:00.159

Part 4: FULL OUTER JOIN

1. Retrieve a list of all customers and suppliers, including those who have no corresponding records in the other table.

Exercise2/postgres@localhost

No limit

Query Query History

```
select c.contact_name,c.customer_id,s.contact_name,s.supplier_id
from customers c full outer join orders o on o.customer_id=c.customer_id
full outer join order_details od on o.order_id=od.order_id
full outer join products p on p.product_id=od.product_id
full outer join suppliers s on s.supplier_id=p.supplier_id;
```

Data Output Messages Notifications

SQL

	contact_name character varying (30)	customer_id character	contact_name character varying (30)	supplier_id smallint
1	Paul Henriot	VINET	Antonio del Valle Saave...	5
2	Paul Henriot	VINET	Chandra Leka	20
3	Paul Henriot	VINET	Elio Rossi	14
4	Karin Josephs	TOMSP	Mayumi Ohno	6
5	Karin Josephs	TOMSP	Wendy Mackenzie	24
6	Mario Pontes	HANAR	Robb Merchant	19
7	Mario Pontes	HANAR	Wendy Mackenzie	24
8	Mario Pontes	HANAR	Shelley Burke	2
9	Mary Saveley	VICTE	Lars Peterson	9
10	Mary Saveley	VICTE	Giovanni Giudici	26
11	Mary Saveley	VICTE	Shelley Burke	2
12	Pascale Cartrain	SUPRD	Peter Wilson	8
13	Pascale Cartrain	SUPRD	Beate Vileid	15
14	Pascale Cartrain	SUPRD	Eliane Noz	28
15	Mario Pontes	HANAR	Elio Rossi	14
16	Mario Pontes	HANAR	Guylène Nodier	18
17	Mario Pontes	HANAR	Anne Heikkonen	23
18	Yang Wang	CHOPS	Carlos Diaz	10
19	Yang Wang	CHOPS	Jean-Guy Lauzon	25
20	Yang Wang	CHOPS	Yoshi Nagase	4
21	Michael Holz	RICSU	Charlotte Cooper	1
22	Michael Holz	RICSU	Ian Devling	7
23	Michael Holz	RICSU	Michael Björn	17

Total rows: 2157 Query complete 00:00:00.115

2. Get a list of all employees and their orders, including employees who haven't processed any orders and orders that don't have an assigned employee.

Exercise2/postgres@localhost

No limit

Query | Query History

```
1 select e.employee_id,e.first_name,o.order_id
2 from employees e full outer join orders o
3 on e.employee_id=o.employee_id;
```

Data Output | Messages | Notifications

	employee_id smallint	first_name character varying (10)	order_id smallint
1	5	Steven	10248
2	6	Michael	10249
3	4	Margaret	10250
4	3	Janet	10251
5	4	Margaret	10252
6	3	Janet	10253
7	5	Steven	10254
8	9	Anne	10255
9	3	Janet	10256
10	4	Margaret	10257
11	1	Nancy	10258
12	4	Margaret	10259
13	4	Margaret	10260
14	4	Margaret	10261
15	8	Laura	10262
16	9	Anne	10263
17	6	Michael	10264
18	2	Andrew	10265
19	3	Janet	10266
20	4	Margaret	10267
21	8	Laura	10268
22	5	Steven	10269
23	1	Nancy	10270

Total rows: 830 | Query complete 00:00:00.105

3. Retrieve all products and their order details, ensuring that products with no orders and orders missing product details are also included.

Exercise2/postgres@localhost

No limit

Query Query History

```
select od.order_id,p.product_id,p.product_name,od.quantity
from products p full outer join order_details od
on p.product_id=od.product_id;
```

Data Output Messages Notifications

	order_id smallint	product_id smallint	product_name character varying (40)	quantity smallint
1	10248	11	Queso Cabrales	12
2	10248	42	Singaporean Hokkien Fried Mee	10
3	10248	72	Mozzarella di Giovanni	5
4	10249	14	Tofu	9
5	10249	51	Manjimup Dried Apples	40
6	10250	41	Jack's New England Clam Cho...	10
7	10250	51	Manjimup Dried Apples	35
8	10250	65	Louisiana Fiery Hot Pepper Sau...	15
9	10251	22	Gustaf's Knäckebröd	6
10	10251	57	Ravioli Angelo	15
11	10251	65	Louisiana Fiery Hot Pepper Sau...	20
12	10252	20	Sir Rodney's Marmalade	40
13	10252	33	Geitost	25
14	10252	60	Camembert Pierrot	40
15	10253	31	Gorgonzola Telino	20
16	10253	39	Chartreuse verte	42
17	10253	49	Maxilaku	40
18	10254	24	Guaraná Fantástica	15
19	10254	55	Pâté chinois	21
20	10254	74	Longlife Tofu	21
21	10255	2	Chang	20
22	10255	16	Pavlova	35
23	10255	36	Inlagd Sill	25

Total rows: 2155 Query complete 00:00:00.118

SELF JOIN & CROSS JOIN

1. Find all pairs of employees who work in the same city (use a SELF JOIN).

Query Query History

```sql
select
    e1.employee_id as emp1_id,
    e1.first_name as emp1_name,
    e2.employee_id as emp2_id,
    e2.first_name as emp2_name,
    e1.city
from employees e1
join employees e2
on e1.city = e2.city
and e1.employee_id < e2.employee_id
```

Data Output Messages Notifications

Data Output Messages Notifications

SQL

	emp1_id smallint	emp1_name character varying (10)	emp2_id smallint	emp2_name character varying (10)	city character varying (15)
1	1	Nancy	8	Laura	Seattle
2	5	Steven	9	Anne	London
3	5	Steven	7	Robert	London
4	5	Steven	6	Michael	London
5	6	Michael	9	Anne	London
6	6	Michael	7	Robert	London
7	7	Robert	9	Anne	London

2. Retrieve a cross join of all categories and suppliers, showing every possible combination.

Query Query History

```
select c.category_id,c.category_name,s.supplier_id,s.company_name
from categories c
cross join suppliers s;
```

Data Output Messages Notifications

	category_id smallint	category_name character varying (15)	supplier_id smallint	company_name character varying (40)
1	1	Beverages	1	Exotic Liquids
2	2	Condiments	1	Exotic Liquids
3	3	Confections	1	Exotic Liquids
4	4	Dairy Products	1	Exotic Liquids
5	5	Grains/Cereals	1	Exotic Liquids
6	6	Meat/Poultry	1	Exotic Liquids
7	7	Produce	1	Exotic Liquids
8	8	Seafood	1	Exotic Liquids
9	1	Beverages	2	New Orleans Cajun Delights
10	2	Condiments	2	New Orleans Cajun Delights
11	3	Confections	2	New Orleans Cajun Delights
12	4	Dairy Products	2	New Orleans Cajun Delights
13	5	Grains/Cereals	2	New Orleans Cajun Delights
14	6	Meat/Poultry	2	New Orleans Cajun Delights
15	7	Produce	2	New Orleans Cajun Delights
16	8	Seafood	2	New Orleans Cajun Delights
17	1	Beverages	3	Grandma Kelly's Homestead
18	2	Condiments	3	Grandma Kelly's Homestead

Total rows: 232 Query complete 00:00:00.108